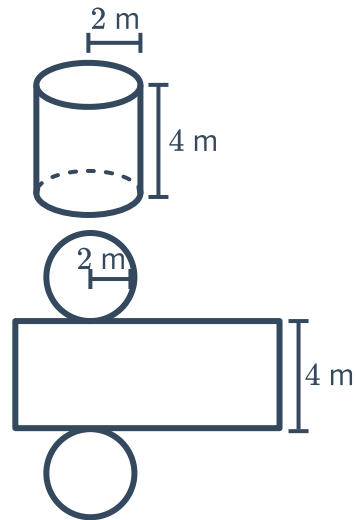


Surface area of cylinders



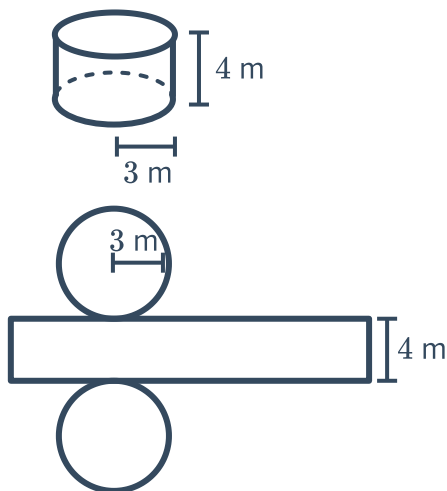
- 1 Consider the following cylinder together with its net:

- Find the circumference of the circular base. Round your answers to two decimal places.
- Hence, find the area of the curved face of the cylinder. Round your answers to two decimal places.

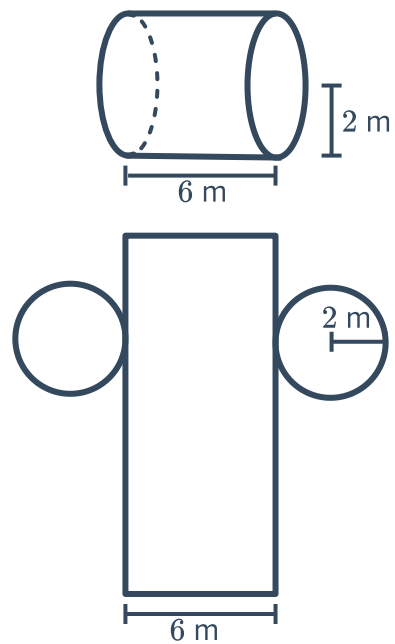


- 2 For each of the following pair of cylinders and their nets, find the curved surface area of the cylinder. Round your answers to two decimal places.

a



b

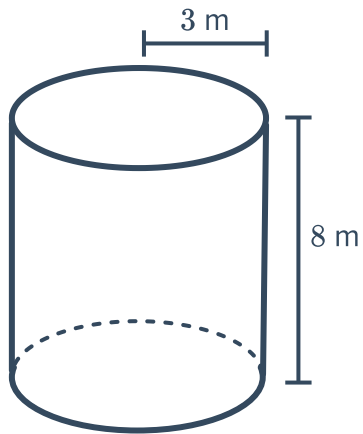


3 For each of the following cylinders:

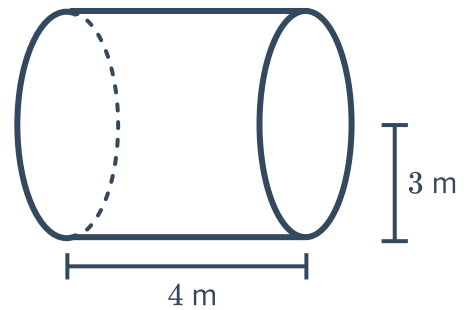


- i Find the curved surface area of the cylinder. Round your answers to two decimal places.
- ii Find the total surface area of the cylinder. Round your answers to two decimal places.

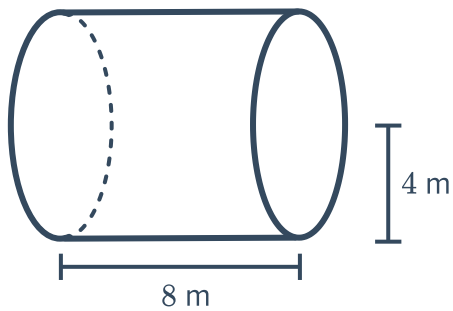
a



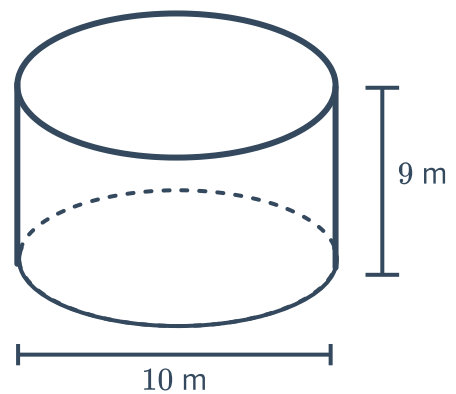
b



c



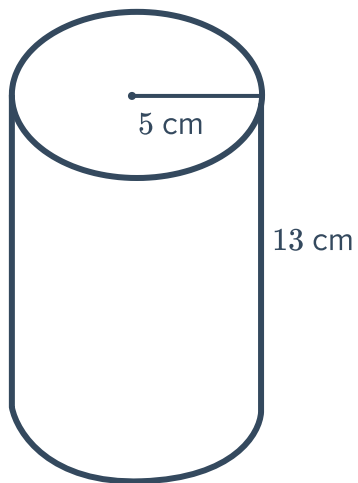
d



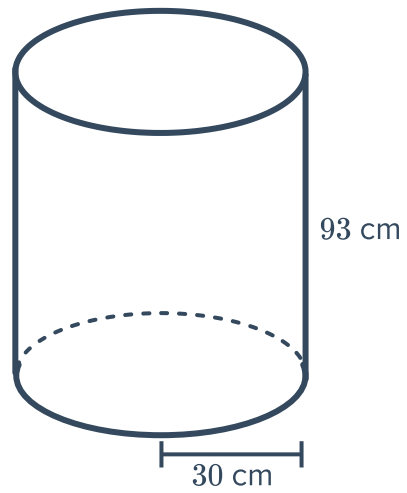
4 Find the surface area of the following cylinders.



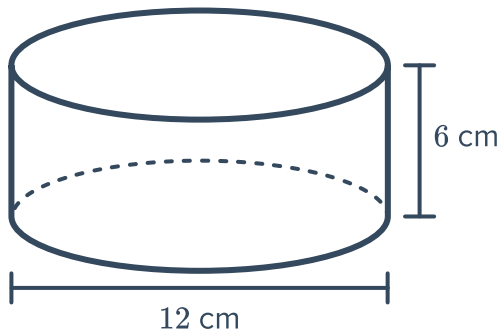
a



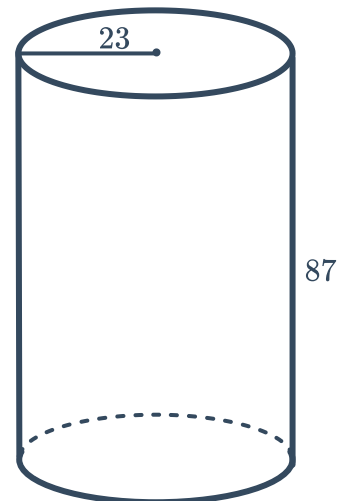
b



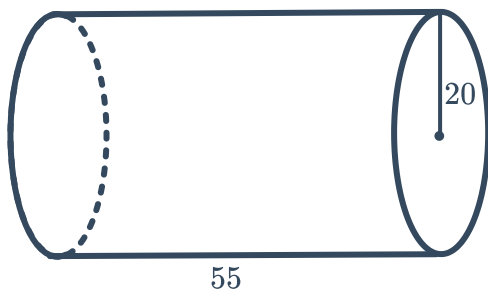
c



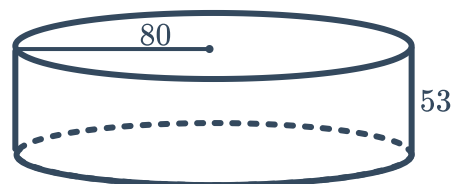
d



e



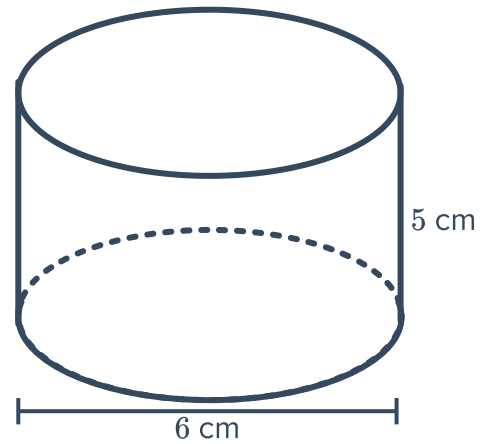
f



5 Consider the following cylinder:



- a Find the surface area of the cylinder to one decimal place.
- b Find the surface area of the cylinder in square millimetres to the nearest whole number.



6 The area of the circular face on a cylinder is $8281\pi \text{ m}^2$. The total surface area of the cylinder is $25\,662\pi \text{ m}^2$.

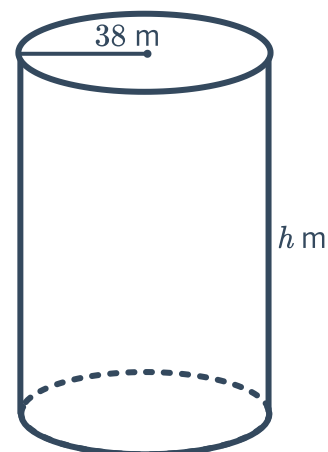
- a Find the radius of the cylinder.
- b Find the height of the cylinder.

7 The area of the circular face on a cylinder is $225\pi \text{ m}^2$. The total surface area of the cylinder is $3210\pi \text{ m}^2$.

- a Find the value of the radius of the cylinder.
- b Find the height of the cylinder.

8 The following cylinder has a surface area of $25\,547 \text{ m}^2$:

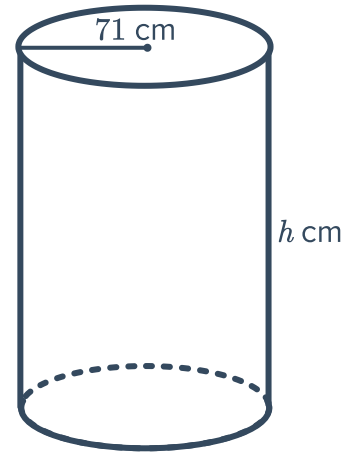
Find the height the cylinder. Round your answer to the nearest whole number.



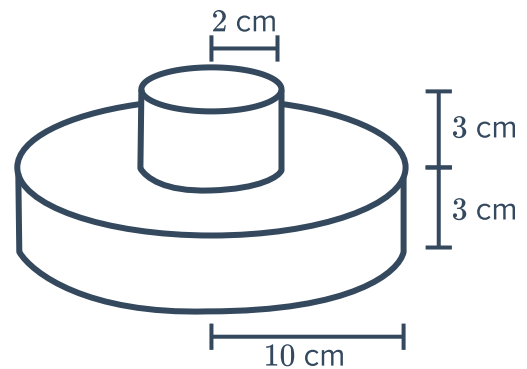
- 9 The following cylinder has a surface area of $54\,425\text{ cm}^2$:



Find the height of the cylinder. Round your answer to the nearest whole number.



- 10 Find the surface area of the following figure.



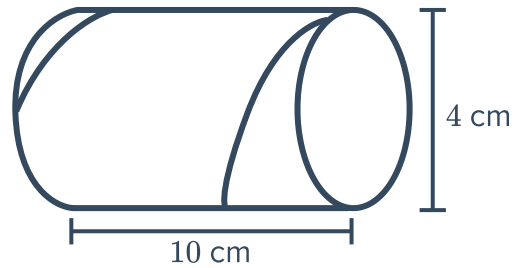
- 11 A cylindrical can of radius 7 cm and height 10 cm is open at one end. What is the external surface area of the can? Round your answer to two decimal places.
- 12 Find the exact surface area of a cylinder with diameter 6 cm and height 21 cm by leaving your answer in terms of π .
- 13 Find the lateral surface area of a cylinder with a radius of 9.2 m and a height of 15.1 m. Round your answer to one decimal places.



Applications

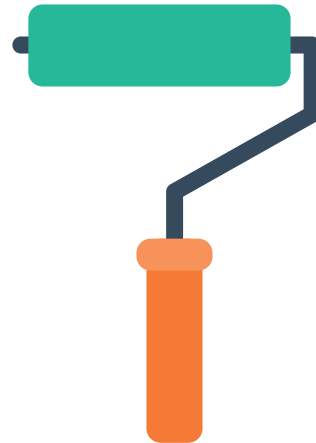
- 14** Paul is using a toilet paper roll for crafts. He has measured the toilet paper roll to have a diameter 4 cm and a length 10 cm.

Find the surface area of the toilet paper roll.
Round your answer to two decimal places.



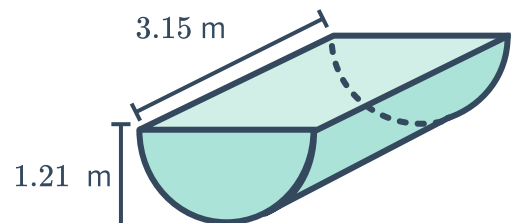
- 15** A paint roller is cylindrical in shape. It has a diameter of 6.2 cm and a width of 38.1 cm.

Find the area painted by the roller when it makes one revolution. Round your answer to two decimal places.



- 16** The diagram shows a water trough in the shape of a half cylinder:

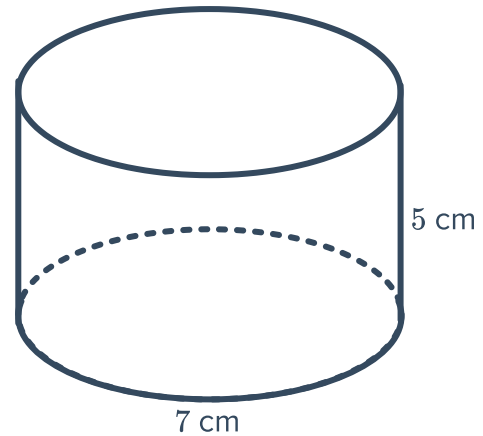
Find the exact surface area of the outside of this water trough, leaving your answer in terms of π .



- 17** Amy and Vincent each have a cylinder. Amy's cylinder has a diameter of 8 cm and a height of 9 cm. Vincent's cylinder has a diameter of 9 cm and a height of 8 cm.
- a** Find the surface area of Amy's cylinder. Round your answer to two decimal places.
 - b** Find the surface area of Vincent's cylinder.
 - c** Which cylinder has a larger surface area? Round your answer to two decimal places.

- 18** Jenny wants to make several cans like the one shown. She plans to cut them out of a sheet of material that has an area of 1683 cm^2 .

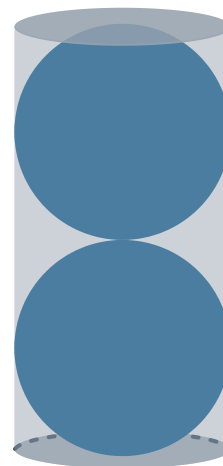
How many complete cans can she make?



- 19** If a spherical ball with a radius of 4.9 m fits exactly inside a cylinder, find the surface area of the cylinder.

- 20** The two identical spherical balls with radii of 2.6 cm fit exactly inside a cylinder.

Find the surface area of the closed cylinder.



- 21 Find the surface area of the brickwork for this silo. Assume that there is a brick roof and no floor.

